

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)			Total percentage of ^{29}Si and ^{30}Si is $100 - 92.2 = 7.8$ % of $^{29}\text{Si} = \frac{2 \times 7.8}{3} = 5.2$ and % of $^{30}\text{Si} = \frac{1 \times 7.8}{3} = 2.6$ (1) $\therefore A_r = \frac{(28 \times 92.2) + (29 \times 5.2) + (30 \times 2.6)}{100}$ (1) $\therefore A_r = \frac{2582 + 151 + 78}{100} = \frac{2811}{100} = 28.1$ (1) Answer only – no mark ecf possible			1		1	
	(b)	(i)		Tetrahedral	1			1		
		(ii)		There are no free electrons or ions to carry the charge	1			1		
		(iii)		Any of the following - There are no electronegativity differences in the Si—Si bond - All the bonding electrons are shared equally between the four Si atoms - Si cannot lose or gain 4 electrons			1	1		

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10.	(c)			M_r HF is 20.01 Solution contains 500g HF in 1000g solution (using $V = m/D$) 1000 g of the solution has a volume of 855 cm^3 or Number of moles of HF in 1000g / 855 cm^3 solution is $\frac{500}{20.01} = 24.98$ (1) 855 cm^3 contain 24.98 mol $\therefore$ Concentration of HF = 29.2 mol dm^{-3} (1) ecf possible						
	(d)	(i)		(There are 6 bonding pairs of electrons and no lone pairs –) position of minimum repulsion taken up (1) Drawing shows clear octahedral shape (1) Bond angle is 90° equatorial / equatorial or 90° equatorial / vertical (accept 180° if vertical bonds only considered) (1)	1					
						2		3		

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10.		(ii)		1 mol of H_2SiF_6 (144g) gives 6 mol of F^- ions ($6 \times 19\text{g}$) = 114g (1) $\therefore$ 114 mg of fluoride ions from 144 mg H_2SiF_6 The increase in fluoride ion concentration needed is $0.76 - 0.15 = 0.61 \text{ mg dm}^{-3}$ (1) Amount of H_2SiF_6 needed is $\frac{144 \times 0.61}{114} = 0.77 \text{ mg}$ (1) Accept alternative method Award (3) for cao			3	3	2	
	(e)			MgSiF_6 M_r 166 $\therefore$ Moles of $\text{MgSiF}_6 = 2.6/166 = 1.566 \times 10^{-2}$ (1) $[\text{H}^+]$ is $\therefore 4 \times 1.566 \times 10^{-2} = 0.06265 \text{ mol dm}^{-3}$ (1) $\text{pH} = -\log_{10} [\text{H}^+] = 1.20$ (1) ecf possible		3		3	2	
Question 10 total					3	9	5	17	8	0